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A Hybrid Metaheuristic Optimization and Machine Learning Framework for Smart Healthcare Based Cardiovascular Disease Prediction

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Abstract Cardiovascular diseases (CVDs), particularly heart attacks, remain among the most serious global public health challenges, causing millions of deaths annually and imposing substantial pressure on healthcare systems worldwide. Early identification of at-risk individuals is therefore essential for timely intervention and improved clinical outcomes. In recent years, machine learning (ML) has emerged as a promising tool for supporting medical decision-making through the analysis of complex clinical and lifestyle data. However, the predictive effectiveness of ML models is often limited by the presence of redundant, irrelevant, or noisy features, which can reduce model accuracy, increase computational complexity, and hinder interpretability. To address these challenges, this study proposed an optimization-driven feature-selection framework to enhance the predictive performance of ML-based heart disease diagnosis systems. The proposed approach systematically integrated metaheuristic optimization algorithms with classical ML classifiers to identify the most informative feature subsets from cardiovascular datasets. Two widely used benchmark datasets, namely the CVD dataset and the Heart Attack dataset, were employed to evaluate the effectiveness of the framework. Six state-of-the-art metaheuristic optimization algorithms were investigated for feature selection, including the Bat Algorithm (BA), Whale Optimization Algorithm (WOA), Jaya Algorithm (JA), Particle Swarm Optimization (PSO), Firefly Algorithm (FA), and Flower Pollination Algorithm (FPA). The optimized feature subsets were subsequently evaluated using three well-established ML classifiers: K-Nearest Neighbors (KNN), Naïve Bayes (NB), and Decision Tree (DT). The experimental results demonstrated that integrating metaheuristic feature selection significantly improved classification performance compared with baseline models trained using the full feature space. Among the evaluated optimization methods, the Jaya Algorithm consistently produced the most effective feature subsets across both datasets and achieved the highest predictive performance. In particular, the JA-based models achieved classification accuracies of up to 98.0% on the CVD dataset and 93.4% on the Heart Attack dataset, outperforming other optimization strategies as well as conventional ML models. To demonstrate practical applicability, the developed predictive model was integrated into a prototype mobile

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application that enabled preliminary heart risk assessment based on user-provided health information. By combining optimized feature selection, ML-based prediction, and a mobile-enabled healthcare interface, the proposed system provided a scalable and accessible solution for early cardiovascular risk screening. Overall, the study demonstrated that metaheuristic-driven feature optimization can significantly enhance the reliability and accuracy of ML-based cardiovascular disease prediction systems, thereby contributing to the advancement of intelligent diagnostic solutions within the emerging Healthcare 4.0 ecosystem.

Keywords: Smart healthcare systems; cardiovascular disease prediction; heart attack risk assessment; feature selection; metaheuristic optimization; machine learning.

1. Introduction

Cardiovascular diseases (CVDs), including heart attacks, remain a leading cause of morbidity and mortality worldwide, accounting for approximately 17.9 million deaths annually according to the World Health Organization (WHO) [1]. The substantial global burden of these conditions underscores the urgent need for effective preventive strategies and timely risk identification. Although significant advances have been achieved in clinical diagnostics and treatment, the early identification of individuals at risk of cardiovascular events remains a persistent challenge. Conventional diagnostic procedures typically depend on comprehensive clinical assessments and specialized medical tests, which are often costly, time-consuming, and highly reliant on expert interpretation. Such limitations may delay timely medical intervention, thereby increasing the likelihood of severe complications and adverse outcomes. Consequently, the development of intelligent, automated, and cost-effective predictive systems capable of supporting early cardiovascular risk assessment has become a major objective in contemporary healthcare research.

In recent years, machine learning (ML) has emerged as a powerful paradigm in medical diagnostics, enabling the analysis of complex and high-dimensional healthcare data for early disease prediction and risk assessment. A broad range of ML algorithms, including Decision Tree (DT), K-Nearest Neighbors (KNN), Naïve Bayes (NB), Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM), and Stochastic Gradient Descent (SGD), has been applied to cardiovascular disease prediction with encouraging results [2]. Despite these advances, the predictive performance and generalization capability of many ML-based diagnostic models remain constrained by the presence of irrelevant, redundant, or noisy attributes in clinical datasets. Such features can negatively affect model learning, resulting in overfitting, increased computational complexity, and reduced interpretability, thereby limiting the reliability of ML-driven decision-support systems in clinical practice.

To address these limitations, feature selection has become a critical preprocessing stage in ML-based medical prediction systems. By identifying the most informative attributes within clinical datasets, feature selection techniques can reduce dimensionality, enhance interpretability, and improve predictive performance. A variety of traditional approaches have been employed for this purpose, including Least Absolute Shrinkage and Selection Operator (LASSO) regression, Analysis of Variance (ANOVA), Multiple Spatially Uniform ReliefF (MultiSURF), variance thresholding, and mutual information methods [3]. Although these techniques can effectively remove irrelevant variables and simplify the feature space, they often encounter difficulties when handling nonlinear relationships and complex high-dimensional search spaces commonly found in healthcare data.

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To overcome these challenges, metaheuristic optimization algorithms have recently received considerable attention for feature-selection tasks. These algorithms are capable of efficiently exploring large solution spaces while maintaining a balance between exploration and exploitation to avoid premature convergence to local optima. Several optimization methods, including the Bat Algorithm (BA), Whale Optimization Algorithm (WOA), Particle Swarm Optimization (PSO), Firefly Algorithm (FA), and Flower Pollination Algorithm (FPA), have demonstrated promising results across diverse optimization and machine learning applications [4–7]. Nevertheless, comprehensive comparative investigations of these metaheuristic techniques in the context of cardiovascular disease prediction remain relatively limited. For contextual reference, Figure 1 illustrates the conventional medical diagnosis workflow, which typically progresses from patient medical history and physical examination to diagnostic testing and final clinical diagnosis.



Figure 1. Overview of the conventional medical diagnosis process.

Although metaheuristic optimization methods have demonstrated promising capabilities in various machine learning and optimization applications, their systematic comparative evaluation for feature selection in cardiovascular disease prediction remains limited. Few studies have systematically investigated the relative effectiveness of multiple metaheuristic algorithms for identifying optimal feature subsets across different medical datasets and classification models. Furthermore, the integration of such optimization-driven predictive frameworks into practical healthcare applications, including real-time decision-support systems, has received comparatively little attention in the existing literature.

Accordingly, the primary objective of this study is to develop an optimization-driven feature-selection framework that enhances the predictive performance of machine learning models for cardiovascular disease risk assessment. By systematically integrating multiple metaheuristic optimization algorithms with widely used machine learning classifiers, this study seeks to identify the most informative feature subsets that improve prediction accuracy, reduce model complexity, and enhance the reliability of automated cardiovascular diagnosis systems. To guide the investigation, this study addresses the following research questions:

- Which metaheuristic optimization algorithm provides the most effective feature subset for cardiovascular disease prediction?
- How does optimization-driven feature selection influence the predictive performance of commonly used machine learning classifiers?
- Can the optimized prediction framework be effectively integrated into a practical healthcare application for real-time cardiovascular risk assessment?

This study was further motivated by the need for a methodologically consistent and practically relevant evaluation framework capable of revealing how different metaheuristic feature-selection strategies influence cardiovascular

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disease prediction performance across datasets and classifiers. More specifically, the motivation of this work arises from three main considerations. First, the growing global prevalence of cardiovascular diseases highlights the urgent need for accurate, accessible, and scalable diagnostic tools, particularly in resource-constrained healthcare environments. Second, although machine learning techniques have shown considerable promise in predicting cardiovascular conditions, their effectiveness is often hindered by redundant and noisy medical attributes, emphasizing the importance of robust and efficient feature-selection strategies. Third, with the rapid advancement of Healthcare 4.0 technologies, there is an increasing demand for intelligent and user-centric healthcare solutions that integrate advanced predictive models with practical deployment platforms, thereby enabling real-time risk assessment and personalized health monitoring.

Although preprocessing operations such as handling missing values, normalizing numerical attributes, and reducing redundant features are standard components of machine learning pipelines and have been widely addressed in previous studies, the contribution of the present work lies beyond the preprocessing stage itself. Specifically, this study develops a systematically controlled comparative framework for cardiovascular disease prediction that evaluates six metaheuristic feature-selection algorithms under identical experimental conditions across two benchmark cardiovascular datasets and multiple machine learning classifiers. This unified design enables a fair and reproducible comparison of different optimization strategies and provides clearer insight into their relative effectiveness for feature selection in cardiovascular disease prediction. In addition, the study extends beyond algorithmic evaluation by integrating the optimized predictive model into an IoT-enabled smart healthcare architecture with mobile application support, thereby demonstrating the practical relevance of the proposed framework.

Based on the identified research gap and study objectives, the main contributions of this work can be summarized as follows:

- A comprehensive comparative framework is developed to evaluate six state-of-the-art metaheuristic optimization algorithms for identifying optimal feature subsets in cardiovascular disease prediction tasks.
- The study demonstrates that optimization-driven feature selection significantly enhances the predictive performance of widely used machine learning models compared with baseline models trained using the full feature space.
- Extensive experimental evaluation reveals that the Jaya Algorithm consistently achieves superior classification performance across multiple datasets and machine learning classifiers.
- To demonstrate practical applicability, the proposed predictive framework is implemented within a prototype mobile application that enables accessible and real-time cardiovascular risk assessment, thereby bridging the gap between machine learning research and real-world healthcare deployment.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and presents the necessary background. Section 3 describes the proposed methodology and the overall framework of the optimization–machine learning model. Section 4 introduces the performance evaluation metrics used to assess the predictive models. Section 5 presents the experimental results and provides a comprehensive discussion of the obtained findings. Section 6 discusses the limitations of the study and outlines potential directions for future work. Finally, Section 7 concludes the paper.

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2. Background and Preliminaries

Cardiovascular diseases remain the leading cause of mortality worldwide, thereby intensifying the need for reliable diagnostic and predictive tools that can support early risk assessment and informed clinical decision-making. Recent advances in artificial intelligence and ML have significantly improved the ability to analyze complex medical data and uncover latent patterns associated with cardiovascular risk. These capabilities have enabled the development of intelligent predictive models that can assist healthcare professionals in early detection and timely intervention. Nevertheless, the effectiveness of ML-based diagnostic systems is often limited by feature redundancy and by the challenge of translating predictive models into practical digital healthcare environments. In this context, the present study investigates optimization-driven feature-selection strategies to enhance cardiovascular disease prediction within smart healthcare applications. Accordingly, this section reviews the relevant literature, introduces the metaheuristic optimization algorithms employed for feature selection, and presents the machine learning classification techniques adopted in the proposed framework.

2.1 Related Literature

Heart disease prediction has attracted substantial attention in recent years owing to the growing availability of clinical data and the expanding capabilities of ML in medical decision-support systems. Within this domain, feature selection has emerged as a crucial component for improving predictive performance, reducing data dimensionality, and enhancing the interpretability of diagnostic models. Numerous studies have investigated the integration of feature-selection techniques with various ML algorithms to identify the most informative clinical attributes associated with cardiovascular risk. These approaches have been applied to a range of diagnostic tasks, including early disease detection, patient risk stratification, and prognosis analysis.

Recent studies published between 2019 and 2025 have examined a wide spectrum of feature-selection strategies for heart disease prediction. These studies can generally be grouped into three categories according to the adopted feature-selection approach. The first category comprises statistical and filter-based methods, such as correlation analysis, chi-square tests, and principal component analysis, which select relevant attributes based on statistical relationships within the data. The second category includes wrapper and hybrid methods that combine statistical filtering with classifier-based evaluation to identify informative feature subsets that improve classification performance. The third category consists of metaheuristic optimization-based approaches, which employ evolutionary or swarm-intelligence algorithms to efficiently explore high-dimensional search spaces and identify near-optimal feature combinations. These approaches have been implemented using a variety of ML classifiers, including SVM, Decision Tree (DT), K-Nearest Neighbor (KNN), Random Forest (RF), and ensemble learning models. The primary objective of these studies has been to improve prediction accuracy while simultaneously reducing computational complexity and increasing the robustness of diagnostic systems.

Table 1 summarizes representative studies reported in the literature that examine the impact of feature-selection techniques on heart disease prediction models. The table highlights the diversity of cardiovascular conditions investigated, the feature-selection strategies employed, the machine learning algorithms used for classification, and the evaluation metrics adopted to assess model performance across different datasets. To preserve readability in the literature review, Table 1 provides a structured overview of prior studies, whereas the quantitative performance comparison between the proposed framework and previously reported approaches is presented separately in Table 7.

Table 1. Summary of recent studies on feature-selection techniques for heart disease prediction, including the framework proposed in this study.

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Reference (Year)	Heart Disease Type	Feature Selection Technique	Machine Learning Model(s)	Application Objective	Evaluation Metrics
[8] (2019)	General heart disease	PSO + ACO	SVM, DT, KNN	Prediction and classification	Accuracy, Precision, Recall, F1-score
[9] (2019)	General heart disease	Genetic Algorithm (GA)	SVM	Early diagnosis	Accuracy
[10] (2019)	Coronary artery disease	Imaging biomarkers	PET/SPECT with ML classifiers	Diagnosis and prognosis	Sensitivity, Specificity
[11] (2020)	CVDs	Not specified	DT, SVM, NB	Classification	Accuracy
[12] (2025)	General heart disease	Hybrid FS (ReliefF + Correlation + GA)	Ensemble (RF, SVM, NB)	Prediction	Accuracy, ROC, F1-score
[13] (2021)	Arrhythmia	Not specified	Ensemble learning methods	Arrhythmia detection	Accuracy, F1-score
[14] (2021)	Chronic heart failure	Not specified	ML classifiers	Diagnosis	Accuracy, AUC
[15] (2021)	Arrhythmia	Marine Predators Algorithm (MPA)	CNN	ECG classification	Accuracy
[16] (2022)	Arrhythmia	Not specified	ML classifiers	Heart disease detection	Accuracy, F1-score
[17] (2022)	General heart disease	Filter + Wrapper	DT, SVM, KNN	Impact analysis	Accuracy
[18] (2022)	Arrhythmia	Improved MPA	CNN	Automatic arrhythmia classification	Accuracy
[19] (2022)	General heart disease	Hybrid FS (PCA + filter methods)	Hybrid ML model	Early detection	Accuracy, Sensitivity
[20] (2023)	General heart disease	Hybrid FS + Ensemble Stacking	RF, SVM, LR	Prediction	Accuracy
[6] (2023)	General heart disease	Metaheuristic FS (GA, PSO, DE, ACO)	ML classifiers	Comparative analysis	Accuracy
[4] (2023)	General heart disease	Multiple FS methods (ReliefF, Chi-square, PCA)	ML classifiers	Feature-selection evaluation	Accuracy
[21] (2024)	General heart disease	Modified Bee Algorithm	ML classifiers	Prediction	Accuracy, Precision, Recall
[22] (2024)	CVDs	Chi-square	Voting Ensemble	Early detection	Accuracy
[23] (2025)	CVDs	Multi-FS + PSO	XGBoost	Prediction	Accuracy, AUC
[40] (2022)	CVDs	Multiple ML-based feature approaches	KNN, NB, DT, SVM	CVD prediction	Accuracy, Precision, Recall
[41] (2023)	CVDs	Improved Hawks Optimizer (HHO)	Neural Network classifiers	CVD prediction	Accuracy, Specificity, Sensitivity, F1-score
[42] (2023)	CVDs	Binary Krill Herd (B-KHA) metaheuristic	CapNet (Deep Learning)	CVD risk prediction	Accuracy, Precision, Recall, F1-score, MCC
Proposed Work (This Study)	Cardiovascular disease and heart attack	Metaheuristic feature selection (BA, WOA, PSO, FA, FPA, JA)	KNN, NB, DT	Cardiovascular risk prediction and real-time assessment	Accuracy, Precision, Recall, F1-score

Several important observations and limitations can be drawn from the studies summarized in Table 1. First, many existing approaches rely on individual feature-selection methods or limited hybrid strategies, which may restrict the exploration of optimal feature subsets in complex cardiovascular datasets. Second, only a limited number of studies provide systematic comparative evaluations of multiple metaheuristic optimization algorithms under identical experimental conditions, making it difficult to determine which methods consistently perform best across different datasets and classifiers. Third, many prior studies evaluate their models using a single dataset or a narrow range of classifiers, thereby limiting the generalizability and robustness of the reported findings. Finally, relatively few studies investigate the integration of optimization-driven prediction models into practical smart healthcare environments capable of supporting real-time cardiovascular risk assessment.

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These limitations reveal the need for a more comprehensive investigation of methodology for feature selection in cardiovascular disease prediction. There remains a need for systematic comparative frameworks capable of evaluating multiple optimization algorithms within a unified experimental setting while examining their influence on different machine learning classifiers across diverse cardiovascular datasets. Addressing this gap can provide deeper insight into the effectiveness of optimization-driven feature-selection strategies and their impact on predictive performance in clinical decision-support systems.

Motivated by these limitations, the present study develops a unified optimization-driven framework that enables a fair comparison of six metaheuristic feature-selection algorithms across two cardiovascular datasets and multiple machine learning classifiers. Beyond comparative algorithmic evaluation, the proposed framework extends toward practical deployment through integration into a prototype mobile-based smart healthcare application designed to support accessible and real-time cardiovascular risk prediction.

2.2 Metaheuristic Optimization Algorithms

Feature selection plays a central role in improving the performance of machine learning–based diagnostic systems by identifying the most informative attributes within clinical datasets while reducing redundancy and dimensionality. In high-dimensional medical data, exhaustive search for the optimal feature subset is computationally impractical. Consequently, metaheuristic optimization algorithms provide an effective alternative by efficiently exploring the search space to identify near-optimal solutions. These algorithms employ adaptive search strategies that balance exploration and exploitation, thereby enabling the discovery of relevant feature combinations capable of improving classification accuracy and model generalization.

In this study, several metaheuristic optimization algorithms are employed to identify the most significant features associated with cardiovascular disease prediction. The selected algorithms include FPA, FA, PSO, JA, WOA, and BA. Each algorithm searches for an optimal feature subset by evaluating candidate solutions according to the predictive performance achieved by the underlying machine learning classifiers.

The proposed framework integrates these optimization algorithms within a machine learning pipeline designed for cardiovascular risk prediction. The workflow begins with data acquisition and preprocessing, followed by optimization-driven feature selection, model training using selected classifiers, and final performance evaluation using standard diagnostic metrics. As illustrated in Figure 2, the framework comprises multiple stages, including preprocessing, feature selection, machine learning classification, and performance evaluation based on metrics such as accuracy, recall, specificity, ROC, and F1-score. The mathematical formulation and operational mechanisms of the employed metaheuristic algorithms are presented in the following subsections.

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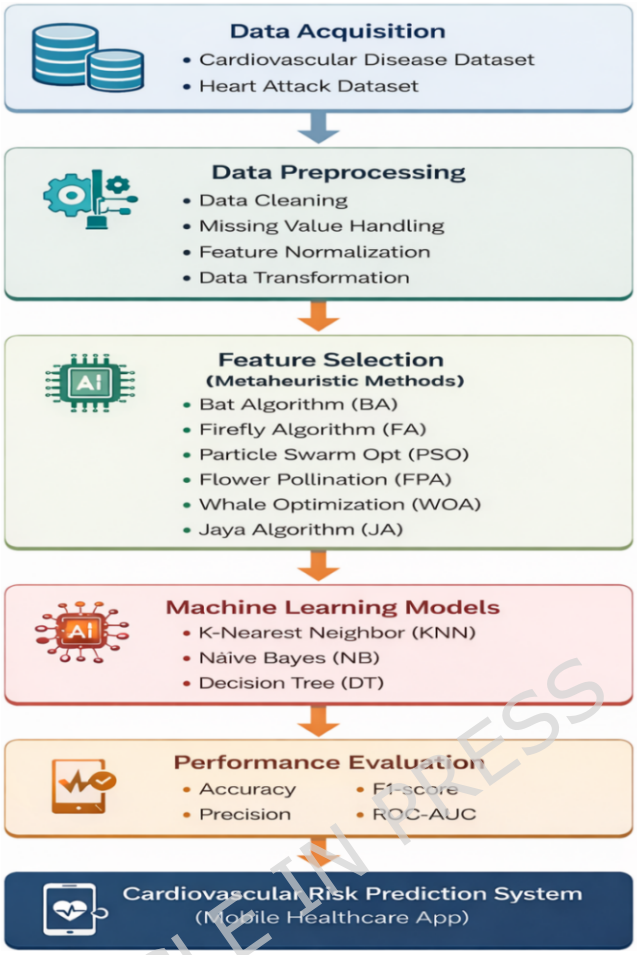


Figure 2. Workflow of the proposed machine learning–based framework for cardiovascular disease prediction.

2.2.1 Flower Pollination Algorithm (FPA)

The Flower Pollination Algorithm (FPA), proposed by Yang, is a nature-inspired metaheuristic optimization method that simulates the pollination process of flowering plants [6]. In this algorithm, candidate solutions are represented as pollen particles, and the search process mimics the natural behavior of pollinators transferring pollen between flowers. FPA is governed by four principal mechanisms: global pollination, local pollination, flower constancy, and a switch probability that regulates the balance between exploration and exploitation during optimization.

Global pollination corresponds to cross-pollination performed by pollinators such as insects or birds, allowing pollen to travel over relatively long distances. In the algorithmic formulation, this process is modeled as a global search mechanism in which candidate solutions are updated using Lévy flight distributions. The update rule for global pollination is expressed as:

$$X_i^{t+1} = X_i^t + \gamma L(\lambda)(g^* - X_i^t) \quad (1)$$

where X_i^t denotes the solution vector of the i-th pollen at iteration t, g^* represents the current best solution in the population, and $L(\lambda)$ denotes the Lévy flight step size that controls the magnitude of the search movement. The Lévy distribution used to model long-distance pollination behavior is defined as:

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$$L\frac{\lambda\Gamma(\lambda)sin(\frac{\pi\lambda}{2})}{\pi}\frac{1}{s^{\lambda+1}},(s>>s_0>0) \quad (2)$$

where $\Gamma(\cdot)$ denotes the gamma function, s represents the step size, and s_0 denotes the minimum threshold value, typically set to a small constant in practical implementations.

Local pollination represents self-pollination or pollination within nearby flowers, corresponding to a local search process in the optimization framework. In this mechanism, candidate solutions are updated using neighboring solutions selected from the same population. The update rule for local pollination is expressed as:

$$X_i^{t+1} = X_i^t + \in (X_j^t - X_k^t) \quad (3)$$

where X_j^t and X_k^t represent two pollen solutions randomly selected from the population, and ϵ is a random number drawn from a uniform distribution. The algorithm employs a switch probability $p \in [0,1]$ to control the transition between global and local pollination processes.

By combining global exploration through Lévy flights with local exploitation through neighborhood-based search, FPA effectively balances diversification and intensification within the solution space. This capability enables the algorithm to identify high-quality feature subsets efficiently, making it well suited for feature-selection tasks in cardiovascular disease prediction models.

2.2.2 Firefly Algorithm (FA)

The FA algorithm is a swarm-based metaheuristic optimization method inspired by the flashing behavior of fireflies, where each firefly represents a candidate solution and its brightness corresponds to the fitness value of that solution. In FA, fireflies are attracted to brighter individuals, and the degree of attractiveness decreases with increasing distance. The attractiveness between two fireflies is defined as:

$$\beta(r) = \beta_0 e^{-\gamma r^2} \quad (4)$$

where β_0 denotes the initial attractiveness, γ represents the light absorption coefficient, and $r_{ij} = \| x_i - x_j \|$ is the Euclidean distance between fireflies i and j. The position of a firefly is updated based on the attractiveness toward brighter fireflies combined with a randomization component, which is expressed as:

$$x_i^{t+1} = x_i^t + \beta_0 e^{-\gamma r_{ij}^2}(x_i^t - x_j^t) + \alpha\varepsilon \quad (5)$$

where α is the randomization parameter controlling stochastic movement, and ϵ represents a random vector drawn from a uniform distribution. This combination of deterministic attraction and stochastic exploration enables FA to effectively explore the search space while reducing the risk of premature convergence to local optima. Owing to its adaptive search capability, FA has been successfully applied to feature selection tasks in medical datasets, including heart disease prediction problems [24].

2.2.3 Particle Swarm Optimization (PSO)

The PSO algorithm is a population-based stochastic optimization technique inspired by the collective behavior of bird flocks and fish schools. In PSO, each candidate solution is represented by a particle that moves through the search space to identify the optimal solution. Each particle is characterized by its position, which represents a potential solution, and its velocity, which determines how the position changes during successive iterations [1].

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During the optimization process, each particle maintains a memory of the best position it has encountered, referred to as the personal best ($pbest$). In addition, particles exchange information with the rest of the swarm, allowing them to move toward the best solution discovered by any particle in the population, known as the global best ($gbest$). This information-sharing mechanism enables PSO to balance exploration of the search space with exploitation of promising solutions.

The velocity of each particle is updated iteratively according to the following equation [1, 4]:

$$v_i^{t+1} = v_i^t + c_1 r_1 (pbest_i^t - p_i^t) + c_2 r_2 (pbest^t - p_i^t) \quad (6)$$

where v_i^{t+1} denotes the velocity of particle i at iteration $t+1$, v_i^t is the velocity at iteration t, x_i^t represents the current position of particle i, $pbest_i$ denotes the best position previously achieved by the particle, and $gbest$ represents the best position identified by the entire swarm. The parameters c_1 and c_2 are acceleration coefficients that control cognitive and social learning components, while r_1 and r_2 are random numbers uniformly distributed within the interval $[0, 1]$.

After updating the velocity, the position of each particle is adjusted according to:

$$p_i^{t+1} = p_i^t + v_i^{t+1} \quad (7)$$

Through successive updates of velocity and position, the swarm collectively converges toward an optimal or near-optimal solution within the search space. Due to its simplicity, computational efficiency, and strong global search capability, PSO has been widely applied in machine learning and engineering optimization problems, including feature selection in cardiovascular disease prediction models.

2.2.4 Jaya Algorithm (JA)

The JA algorithm is a population-based metaheuristic optimization technique characterized by its simplicity and absence of algorithm-specific control parameters. Unlike many evolutionary algorithms that require careful tuning of parameters, JA iteratively improves candidate solutions by simultaneously moving them toward the best solution and away from the worst solution within the current population [25]. This strategy promotes convergence toward optimal regions of the search space while avoiding poorly performing solutions.

The updated rule governing the evolution of candidate solutions in JA is expressed as:

$$x_{i,j}^{t+1} = x_{i,j}^t + r_1 (x_{best,j}^t - |x_{i,j}^t|) - r_2 (x_{worst,j}^t - |x_{i,j}^t|) \quad (8)$$

where $x_{i,j}^t$ represents the value of the j-th variable of the i-th candidate solution at iteration t, while $x_{best,j}^t$ and $x_{worst,j}^t$ denote the best and worst solutions in the population, respectively. The parameters r_1 and r_2 are random numbers drawn from a uniform distribution $U(0,1)$, which introduce stochastic variation into the search process.

By eliminating algorithm-specific parameters and relying on a direct improvement strategy, JA provides a robust and computationally efficient optimization mechanism. This simplicity makes it particularly suitable for feature selection problems, where the objective is to identify optimal subsets of predictive attributes for improving classification performance in cardiovascular disease prediction models.

2.2.5 Whale Optimization Algorithm (WOA)

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The WOA algorithm, introduced by Mirjalili and Lewis, is a population-based metaheuristic inspired by the bubble-net hunting strategy of humpback whales [6]. In this algorithm, candidate solutions represent search agents that update their positions in the search space to approximate the global optimum. WOA models three main behavioral mechanisms: encircling prey, bubble-net attacking, and global exploration.

(a) Encircling Prey

Humpback whales identify the location of prey and gradually surround it. In the optimization context, the algorithm assumes that the best solution obtained so far approximates the prey location. Other search agents update their positions relative to this best solution according to:

$$\vec{D} = \left| C\vec{X^*}(t) - \vec{X}(t) \right| \quad (9)$$

$$\vec{X}(t+1) = \left| \vec{X^*}(t) - \vec{A} \cdot \vec{D} \right| \quad (10)$$

where $X(t)$ denotes the position vector at iteration t, $X^*(t)$ represents the best solution found so far, and D indicates the distance between the search agent and the prey. The coefficient vectors A and C are defined as:

$$\vec{A} = 2\vec{a} \cdot \vec{r} - \vec{a} \quad (11)$$

$$\vec{C} = 2 \cdot \vec{r} \quad (12)$$

where r is a random vector in the range $[0, 1]$, and the parameter $a$decreases linearly from 2 to 0 throughout the optimization process to balance exploration and exploitation.

(b) Bubble-Net Attacking Mechanism

WOA simulates bubble-net hunting behavior using two strategies: shrinking encirclement and spiral updating. The shrinking mechanism gradually reduces the value of a, causing the search agents to converge toward the best solution. The spiral updating mechanism models the whale's spiral-shaped movement around prey, which is expressed as:

$$\vec{D'} = \left| \vec{X^*}(t) - \vec{X}(t) \right| \quad (13)$$

$$\vec{X}(t+1) = \vec{D'} \cdot e^{bl} cos(2\pi l) + \vec{X^*}(t) \quad (14)$$

where b is a constant defining the spiral shape and l is a random number within the interval $[-1,1]$. At each iteration, search agents probabilistically select between the shrinking encirclement and spiral updating strategies using a probability parameter p:

$$\vec{X}(t+1) = \begin{cases} \vec{X^*}(t) - \vec{A} \cdot \vec{D}, & p < 0.5 \\ \vec{D'} \cdot e^{bl} cos(2\pi l) + \vec{X^*}(t), & p \geq 0.5 \end{cases} \quad (15)$$

(c) Exploration Phase (Search for Prey)

To enhance global search capability, WOA also allows search agents to update their positions relative to randomly selected individuals in the population. When $|A| > 1$, the algorithm favors exploration by moving search agents away

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from the current best solution, enabling a broader search of the solution space. Conversely, when $|A| < 1$, exploitation dominates, and the agents converge toward promising solutions.

Through the combination of these mechanisms, WOA effectively balances exploration and exploitation, enabling efficient search of complex optimization spaces. This capability makes WOA particularly suitable for feature selection problems in cardiovascular disease prediction.

2.2.6 Bat Algorithm (BA)

The BA algorithm is a bio-inspired metaheuristic optimization technique that mimics the echolocation behavior of microbats during prey detection and navigation [26]. In this framework, each bat represents a candidate solution characterized by its position x_i, velocity v_i, frequency f_i, loudness A_i, and pulse emission rate r_i. The algorithm iteratively updates these parameters to explore the search space and identify optimal solutions.

The frequency of each bat is updated as:

$$f_i = f_{\min} + (f_{\max} - f_{\min}) \cdot \beta \quad (16)$$

where β is a random number drawn from a uniform distribution in the interval $[0, 1]$. Based on the updated frequency, the velocity and position of each bat are modified according to:

$$v_i^t = v_i^{t-1} + (x_i^{t-1} - x_*) f_i \quad (17)$$

$$x_i^t = x_i^{t-1} + v_i^t \quad (18)$$

where x^* denotes the current global best solution. In addition to global exploration, BA incorporates local exploitation through random walks around promising solutions. During the search process, the loudness A_i gradually decreases while the pulse emission rate r_i increases, enabling the algorithm to transition from exploration to exploitation.

This adaptive balance between global and local search allows BA to efficiently explore complex optimization spaces. Consequently, BA has been widely applied in feature selection tasks, including medical data analysis and cardiovascular disease prediction.

2.3 Classification and Modeling

This section describes the ML classification methods employed in the proposed predictive framework for cardiovascular disease diagnosis. Initially, multiple classifiers were trained and evaluated to determine their suitability for the classification task. Based on comparative performance analysis, the three most effective models were selected for further experimentation. The primary objective of the classification stage is to distinguish between patients diagnosed with cardiovascular disease and healthy individuals by learning patterns embedded in clinical data. ML algorithms are particularly suitable for this task because they can automatically extract predictive relationships from historical datasets and generalize these patterns to unseen cases. Among the various classification techniques available, this study focuses on three well-established supervised learning NB, KNN, and DT algorithms. These classifiers were selected due to their complementary characteristics, computational efficiency, and strong performance in medical prediction tasks.

2.3.1 K-Nearest Neighbor (KNN)

The KNN algorithm is a widely used supervised learning technique for classification and prediction tasks. It belongs to the class of instance-based learning methods, in which the training data are stored, and classification decisions are made only when a new sample is introduced. In this approach, each data instance is represented as a

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point in a multidimensional feature space, and classification is determined based on its similarity to previously labeled samples [5, 27].

The core assumption of KNN is that data points with similar characteristics are likely to belong to the same class. To classify a new instance, the algorithm computes its distance from all training samples and identifies the nearest K neighbors. The class label is then determined by majority voting among these neighbors. When dealing with continuous prediction tasks, the algorithm instead computes the average value of the neighboring samples. The distance between two samples $X_1 = (x_{11}, x_{12}, \ldots, x_{1n})$ and $X_2 = (x_{21}, x_{22}, \ldots, x_{2n})$ is typically calculated using the Euclidean distance metric:

$$dist(X_1, X_2) = \sqrt{\sum_{i=1}^{n} (x_{1i} - x_{2i})^2} \quad (19)$$

Selecting an appropriate value of K is crucial for achieving optimal performance. Small values of K may cause the classifier to overfit the training data, while excessively large values can oversmooth the decision boundary and reduce classification accuracy. A common strategy for determining the optimal value of K is the elbow method, which evaluates the classification error across different K values and selects the point beyond which further increases produce minimal improvement in performance [27].

2.3.2 Naive Bayes (NB)

The NB classifier is a probabilistic learning method based on Bayes' theorem, which assumes conditional independence among input features [4, 27]. Despite this simplifying assumption, NB has demonstrated strong performance in many classification tasks, particularly when dealing with high-dimensional datasets.

Given a feature vector x, the posterior probability of a class y is computed as:

$$p(y \mid x) = \frac{p(x \mid y) p(y)}{p(x)} \quad (20)$$

The predicted class label corresponds to the class with the highest posterior probability. Due to its computational efficiency, scalability, and robustness when handling large feature spaces, NB has been widely applied in medical diagnostic systems, including cardiovascular disease prediction.

2.3.3 Decision Tree (DT)

The DT algorithm is a supervised learning method that performs classification by recursively partitioning the dataset into smaller subsets based on feature values. The resulting model has a tree-like structure in which internal nodes represent decision rules, branches represent possible outcomes, and leaf nodes correspond to predicted class labels.

The construction of a decision tree is typically guided by impurity measures such as Information Gain or the Gini Index [7,19]. One commonly used measure is entropy, which quantifies uncertainty in the dataset and is defined as:

$$I = -\sum_{C} P(c) \log_2 P(c) \quad (21)$$

where $P(c)$ represents the probability of class c. The Information Gain of an attribute A is then calculated as:

$$Gain(A) = I - I_{res}(A) \quad (22)$$

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where $I_{res}(A)$ denotes the residual entropy after splitting the dataset according to attribute A. The attribute that maximizes the Information Gain is selected as the splitting criterion.

A potential limitation of decision trees is their tendency to overfit when the tree grows excessively deep and captures noise in the training data. To address this issue, several regularization strategies are commonly applied, including pruning, limiting tree depth, and imposing minimum sample thresholds for node splitting. In this study, the DT classifier was further optimized through hyperparameter tuning to improve generalization performance and achieve balanced classification accuracy [28].

3. Proposed Methodology

This study proposes a hybrid optimization–machine learning (OT–ML) framework for cardiovascular disease prediction based on symptom-driven clinical data. The framework integrates metaheuristic optimization techniques with machine learning classifiers to improve feature selection and enhance diagnostic accuracy. By combining optimization-driven feature selection with predictive modeling, the proposed approach seeks to identify the most informative symptom attributes while reducing redundant or irrelevant features that may adversely affect classification performance.

In the proposed system, patient symptom information is initially collected through a mobile-based application designed for real-time health data acquisition. The collected symptom data include indicators commonly associated with cardiovascular disorders, such as chest pain, shortness of breath, cold sweating, dizziness, heartburn, nausea, and chest discomfort. These symptoms represent clinically relevant indicators that can assist in the early identification of heart attack risk. Figure 3 illustrates the major symptoms associated with heart attack that are considered during the data acquisition stage.

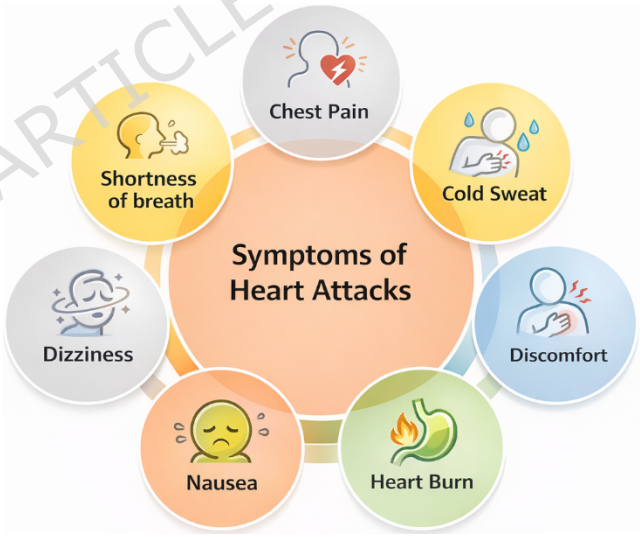


Figure 3. Major symptoms associated with heart attack.

After data acquisition, the collected datasets undergo a sequence of processing stages within the proposed OT–ML framework, as illustrated in Figure 4. The framework comprises several interconnected components that collectively enable efficient cardiovascular disease prediction and risk assessment, as summarized in Algorithm 1.

Step 1: Data Acquisition and Preprocessing

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In the first stage, symptoms and clinical data are collected and stored through a mobile healthcare application connected to the proposed diagnostic system. The acquired datasets are obtained from publicly available medical repositories and include demographic, physiological, and lifestyle-related attributes associated with cardiovascular risk. Since the datasets used in this study are publicly available and anonymized, no personally identifiable information was accessed or processed.

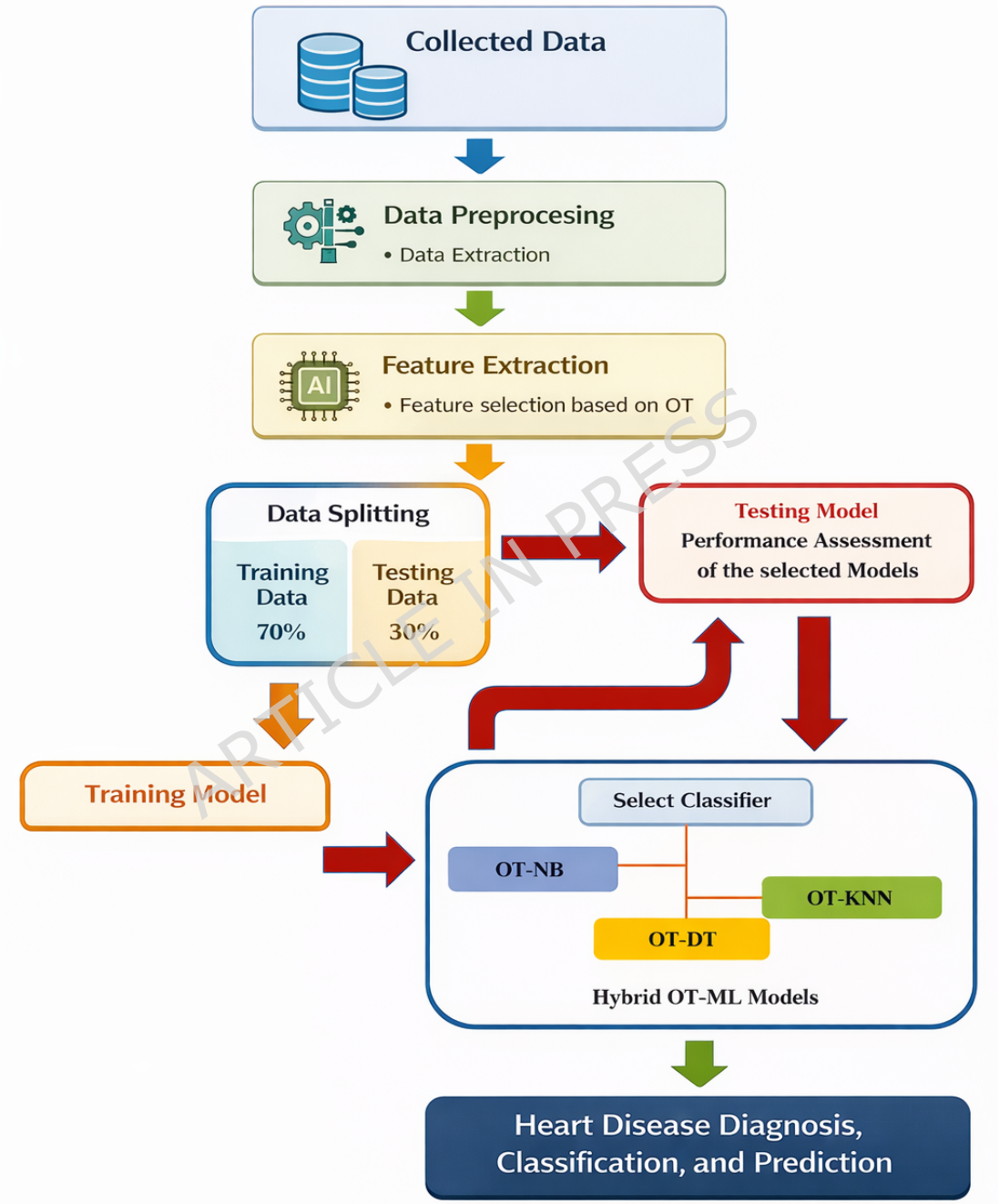


Figure 4. Proposed heart disease diagnosis through the hybrid OT–ML framework integrated with a smart IoT-based cloud system.

Before model training, the collected data undergo several preprocessing procedures to ensure data quality, consistency, and suitability for machine learning analysis. This stage includes data cleaning, missing-value handling,

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normalization of numerical attributes, and transformation of categorical variables when required. Missing values were carefully examined and addressed using appropriate imputation strategies to preserve dataset integrity. Continuous variables such as age, blood pressure, height, and weight were normalized using min–max scaling to ensure comparable feature ranges and to prevent attributes with large numerical magnitudes from dominating the learning process. In addition, categorical attributes such as gender, smoking status, alcohol consumption, and physical activity indicators were encoded into numerical representations suitable for machine learning algorithms. These preprocessing steps were applied consistently across the datasets to ensure comparability and reproducibility. After preprocessing, the cleaned datasets were used as inputs to the metaheuristic optimization algorithms for feature selection and subsequent machine learning classification.

The preprocessing pipeline applied to both datasets followed a systematic sequence of five steps: (1) data cleaning, involving the identification and removal of duplicate, inconsistent, or out-of-range records to ensure dataset integrity; (2) missing-value handling, in which incomplete entries were carefully examined and addressed using appropriate imputation strategies; (3) normalization of continuous variables, including age, height, weight, systolic blood pressure, and diastolic blood pressure, using min–max scaling to map all numerical attributes to a comparable range and prevent features with large magnitudes from dominating the learning process; (4) categorical encoding, whereby nominal attributes such as gender, smoking status, alcohol consumption, and physical activity level were transformed into numerical representations suitable for machine learning processing; and (5) consistent application of all the above procedures across both the CVD dataset and the Heart Attack dataset to ensure that experimental results remain comparable and reproducible across different data sources.

Step 2: Optimization-Based Feature Selection

Following preprocessing, feature extraction is performed to generate informative representations of the collected symptom data. Because medical datasets often contain redundant or irrelevant attributes, an optimization-driven feature selection stage is incorporated to identify the most informative features that contribute to accurate disease prediction. It is important to note that the proposed framework does not employ handcrafted statistical feature extraction. Rather than deriving engineered descriptors from raw signals, the framework operates directly on the original clinical attributes available in the benchmark cardiovascular datasets following the preprocessing stage. The metaheuristic optimization algorithms are then applied to identify the most informative subset from these original features, thereby reducing redundancy while preserving clinically relevant information.

In this study, six metaheuristic optimization algorithms, namely JA, PSO, BA, FPA, FA, and WOA, are employed to search for optimal feature subsets. These algorithms were selected because they represent diverse optimization paradigms, including swarm-intelligence, nature-inspired, and population-based search strategies, and have demonstrated effectiveness in feature-selection tasks in both machine learning and healthcare-related applications. Each algorithm explores the feature space to identify informative feature subsets that maximize classification performance while minimizing redundancy. Their inclusion enables a systematic comparative evaluation of different optimization strategies under identical experimental conditions and facilitates the identification of the most effective approach for cardiovascular disease prediction.

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Let the original feature space be represented as $F = \{f_1, f_2, \ldots, f_n\}$. The objective of the optimization-based feature selection process is to identify an optimal subset $S \subseteq F$ that maximizes classification performance while minimizing feature redundancy. The optimization problem can be formulated as:

$$\text{maximize Fitness}(S) = \alpha \times \text{Accuracy}(S) - \beta \times (|S| / |F|),$$

where $Accuracy(S)$ denotes the classification accuracy obtained using the selected feature subset S, $|S|$ represents the number of selected features, and $|F|$ denotes the total number of original features. The parameters α and β are weighting coefficients that balance classification accuracy and feature dimensionality reduction.

Algorithm 1: Hybrid OT–ML Prediction Framework

Input: Cardiovascular dataset *D*

Output: Predicted cardiovascular disease risk

1. Acquire patient symptom data from dataset *D*.
2. Perform data preprocessing and normalization.
3. Apply metaheuristic optimization algorithms for feature selection.
4. Generate the optimized feature subset *S*.
5. Split the dataset into training and testing sets using a 70:30 ratio.
6. Train the classifiers (NB, DT, and KNN) using the selected feature subset.
7. Evaluate the classifiers using standard performance metrics.
8. Select the best-performing model.
9. Deploy the optimized model within the cloud-based healthcare framework

Step 3: Machine Learning Classification

Following the feature-selection stage, the optimized feature subsets are divided into training and testing sets using a 70:30 partition strategy. This split ratio provides sufficient samples for model training while preserving an independent test set for unbiased evaluation of predictive performance. The selected features are then used to train multiple machine learning classifiers for disease prediction and risk classification.

Three widely used supervised learning models, namely NB, KNN, and DT, are employed for classification. These classifiers were selected because of their effectiveness in medical data classification, computational efficiency, and ability to handle heterogeneous feature spaces. Through training, the classifiers learn predictive relationships between symptom features and disease labels, enabling the system to distinguish between healthy individuals and patients at risk of cardiovascular disease.

Step 4: Model Evaluation and Performance Analysis

To evaluate the effectiveness of the proposed hybrid OT–ML framework, the trained models are tested using multiple cardiovascular datasets. Performance evaluation is conducted using standard classification metrics, including accuracy, precision, recall, F1-score, and receiver operating characteristic (ROC) analysis. Collectively, these metrics assess predictive performance in terms of overall correctness, detection capability, and the balance between false positive and false negative predictions. The evaluation process compares the predictive performance of different classifiers when combined with various metaheuristic feature-selection algorithms, thereby enabling identification of the most effective optimization–classification combination. All experiments were implemented using Python-based machine learning libraries, and the optimization algorithms were executed under identical experimental conditions to ensure a fair comparison among the investigated feature-selection strategies.

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To improve the transparency and reproducibility of the experiments, the hyperparameter settings and optimization configurations used in the proposed framework are explicitly reported in Table 2. The parameter values reported in Table 2 were selected based on commonly adopted configurations in the metaheuristic optimization literature. For the Flower Pollination Algorithm, the switch probability was set to 0.8 to balance global and local search. For the Firefly Algorithm, the randomization parameter α=0.2 and light absorption coefficient γ=1 were used to control stochastic movement. For PSO, an inertia weight of w=0.7 with cognitive and social coefficients of c_1=c_2=1.5 was adopted. The Whale Optimization Algorithm used a linearly decreasing control parameter a from 2 to 0. The Bat Algorithm employed a loudness of 0.5 and pulse rate of 0.5 within a frequency range of [0, 2]. These settings were maintained consistently across all experiments to ensure comparability of results.

The table summarizes the parameter settings adopted for both the investigated metaheuristic optimization algorithms and the machine learning classifiers. These settings were selected according to commonly used configurations reported in the metaheuristic optimization and machine learning literature and were kept fixed across all experiments to ensure a fair and controlled comparison among the evaluated methods. Such explicit reporting of the experimental configuration facilitates reproducibility and provides a clearer basis for interpreting the obtained performance results.

Table 2. Experimental hyperparameter settings and optimization configurations used in the proposed framework.

Parameters	Values
Population size	5
Iterations	100
KNN	$K = 3$
DT	criterion = 'gini', splitter = 'best', max_depth = None, min_samples_split = 2, min_samples_leaf = 1
NB	Default
FPA	Switch probability $(p) = 0.8$, Lévy flight parameter = default
FA	Randomization parameter $(\alpha) = 0.2$, attraction coefficient $(\beta_0) = 1$, light absorption coefficient $(\gamma) = 1$
PSO	Inertia weight $(w) = 0.7$, cognitive coefficient $(c_1) = 1.5$, social coefficient $(c_2) = 1.5$
WOA	Control parameter (a)linearly decreased from 2 to 0
BA	Loudness $(A) = 0.5$, pulse rate $(r) = 0.5$, frequency range = $[0, 2]$

The reported parameter settings were applied consistently across all experiments to maintain fairness of comparison among the evaluated algorithms while providing a balanced trade-off between computational efficiency and optimization effectiveness.

Step 5: IoT-Based Cloud Deployment

The proposed predictive framework is ultimately deployed within a smart healthcare environment supported by Internet of Things (IoT) and cloud computing technologies. In this stage, the optimized predictive models are integrated into a cloud-based healthcare platform that enables real-time cardiovascular risk prediction through mobile devices. This deployment allows patients to submit symptom information through a mobile application, while healthcare professionals can monitor patient data and prediction outcomes through a centralized medical dashboard. The integration of IoT and cloud infrastructure therefore supports scalable, secure, and real-time healthcare decision-making. This three-stage deployment architecture demonstrates the end-to-end feasibility of the proposed framework within a real-world IoT-enabled smart healthcare environment, supporting its applicability beyond experimental evaluation.

To enable practical deployment of the proposed cardiovascular disease prediction system, a smart healthcare architecture integrating IoT, cloud computing, and artificial intelligence technologies is developed [39]. The objective

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of this framework is to provide real-time disease prediction and monitoring by allowing patients to submit symptom data through mobile devices while enabling healthcare professionals to access diagnostic results remotely. The integration of IoT devices with cloud-based predictive models supports continuous monitoring, efficient data processing, and scalable healthcare services.

The complete workflow of the proposed hybrid OT–ML diagnostic framework is illustrated in Figure 4, which depicts the integration of the optimization-driven machine learning pipeline within a cloud-enabled IoT healthcare environment. The framework comprises several computational stages, including data collection, preprocessing, feature extraction, optimization-based feature selection, machine learning classification, and diagnostic prediction. In this architecture, patient medical data are first collected and preprocessed before being analyzed by the proposed hybrid optimization–machine learning models. Feature selection is performed using metaheuristic optimization algorithms to identify the most relevant symptom attributes. The optimized feature subsets are then used to train machine learning classifiers, including Naïve Bayes, Decision Tree, and K-Nearest Neighbor models. These classifiers are integrated with optimization techniques to form hybrid prediction models (OT-NB, OT-DT, and OT-KNN), which are evaluated to determine the most effective configuration for cardiovascular disease prediction.

Following model training, the system performs performance assessment using standard evaluation metrics to identify the most accurate predictive model. The optimized classification model is subsequently deployed within the cloud environment to support real-time diagnostic services. The final output of the system provides automated heart disease diagnosis, classification, and risk prediction based on the submitted patient symptoms.

Although the proposed framework was evaluated using two different data sources, namely a real-world cardiovascular disease dataset and a synthetic heart attack dataset, the generalizability of the results should be interpreted with appropriate caution. The inclusion of both datasets provides an initial indication of the framework's ability to operate across different data characteristics. However, the synthetic heart attack dataset may not fully capture the complexity, variability, and noise patterns typically observed in real-world clinical populations. Therefore, although the developed prediction framework was integrated into a prototype IoT-enabled smart healthcare environment to demonstrate practical applicability, further validation using larger real-world and multi-center cardiovascular datasets remains necessary to more comprehensively assess the reliability, robustness, and generalizability of the proposed framework across diverse healthcare settings.

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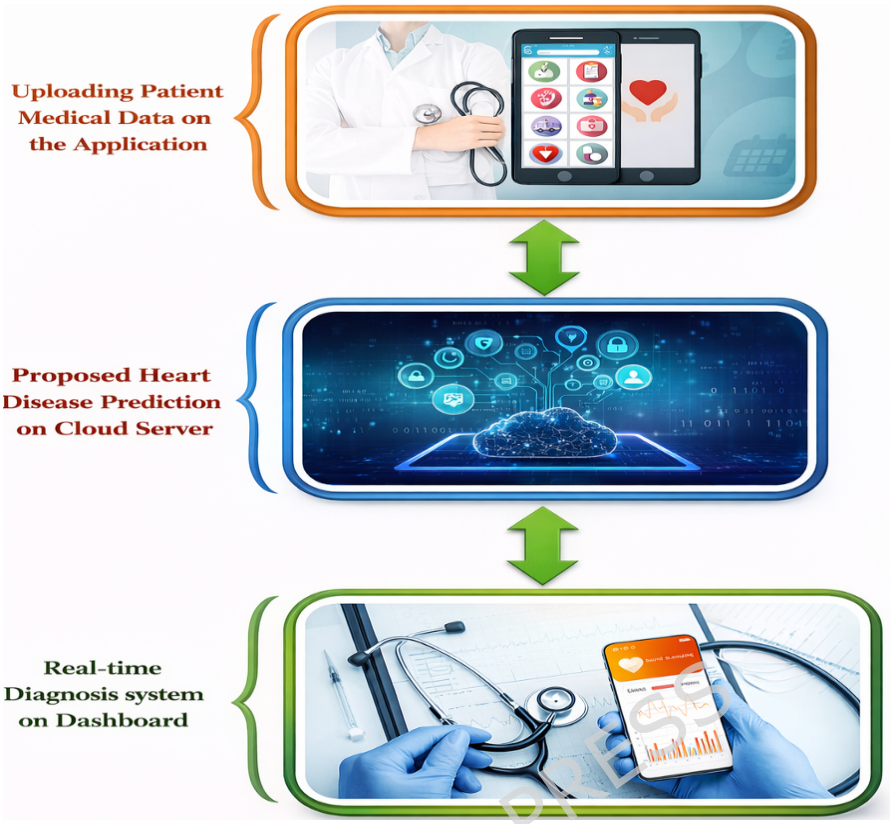


Figure 5. Proposed cloud-based framework through the IoT platform for real-time heart disease diagnosis.

To facilitate real-world implementation, the real-time deployment architecture of the proposed healthcare system is illustrated in Figure 5. The IoT-based healthcare architecture comprises three main operational stages. Although the inclusion of a synthetic heart attack dataset provides useful experimental diversity, it does not substitute for validation using real-world clinical datasets. Accordingly, further assessment on large-scale clinical data remains necessary to comprehensively evaluate the robustness and generalizability of the proposed framework in practical healthcare environments.

Stage 1: Patient Data Acquisition via Mobile Application

In the first stage, patients interact with the healthcare system through a mobile application that allows them to enter their medical symptoms and personal health information. The mobile application functions as an IoT interface for collecting real-time patient data, including symptoms related to cardiovascular disorders such as chest pain, dizziness, shortness of breath, nausea, heartburn, and cold sweating. Once the data are submitted, they are securely transmitted to the cloud server for further processing and analysis. This stage enables continuous patient interaction with the diagnostic system and provides a convenient mechanism for remote health-data collection without requiring direct hospital visits.

Stage 2: Cloud-Based Processing and Disease Prediction

In the second stage, the collected patient data are transmitted to the cloud server, where the proposed hybrid OT–ML prediction model is deployed for automated cardiovascular risk analysis. The cloud infrastructure performs several tasks, including data storage, feature processing, model inference, and diagnostic prediction. The cloud environment provides substantial computational resources that enable efficient execution of the optimization-based feature-

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selection algorithms and machine learning classifiers. The optimized predictive model analyzes the incoming patient data and generates cardiovascular risk predictions in real time. In addition, cloud computing ensures scalability, allowing the system to support many users while maintaining high computational efficiency. Centralized cloud storage also supports secure data management by preserving patient records with high reliability and reducing the risk of unauthorized access or data manipulation.

Stage 3: Clinical Monitoring and Decision-Support Dashboard

In the final stage, the prediction results generated by the cloud-based system are delivered to healthcare professionals through a web-based monitoring dashboard. This dashboard enables clinicians to review patient medical records, examine predicted cardiovascular risk levels, and monitor health trends over time. The dashboard interface supports medical decision-making by providing visual representations of patient data and diagnostic outcomes. Physicians can use this information to evaluate patient conditions, recommend preventive measures, or initiate additional medical examinations when necessary. By providing real-time access to predictive analytics, the system assists healthcare providers in delivering timely and informed clinical interventions.

The proposed framework offers several advantages for intelligent healthcare systems. First, it enables real-time health monitoring through mobile-based patient data acquisition. Second, the cloud computing infrastructure provides scalable computational resources capable of efficiently processing large volumes of medical data. Third, the integration of optimization-based feature selection improves diagnostic accuracy by identifying the most relevant clinical attributes while reducing redundant features. In addition, the system supports remote healthcare accessibility, allowing patients to obtain predictive health assessments without visiting medical facilities. Finally, centralized dashboards enhance clinical decision-making by enabling healthcare professionals to monitor patient conditions through real-time predictive analytics.

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4. Performance Metrics

The effectiveness of the proposed hybrid OT–ML framework is evaluated using several standard classification performance metrics. These metrics are employed to assess the predictive capability of the proposed framework and to compare its performance with conventional machine learning models that do not incorporate optimization-based feature selection. The evaluation is conducted on the considered cardiovascular disease datasets to determine how effectively the integration of optimization techniques improves classification performance.

Model performance is assessed by analyzing the agreement between the predicted class labels generated by the classifier and the corresponding actual labels in the dataset. This evaluation is based on the confusion matrix, which summarizes the prediction outcomes of the classification model. The confusion matrix consists of four fundamental components: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

TP represents instances of cardiovascular disease that are correctly identified by the model. TN corresponds to healthy cases that are correctly classified as non-disease instances. FP indicates healthy cases that are incorrectly classified as disease cases, whereas FN represents disease cases that are incorrectly predicted as healthy. These four values form the basis for calculating several evaluation metrics that quantify the classification performance of the predictive models.

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Table 3 summarizes the performance metrics used to evaluate the proposed framework. These metrics provide complementary perspectives on classification performance, enabling a comprehensive assessment of the predictive capability of the proposed OT–ML model.

5. Experimental Results and Discussion

This section presents the experimental evaluation of the proposed hybrid optimization–machine learning framework for cardiovascular disease prediction. The performance of the framework was assessed using two datasets related to cardiovascular disease and heart attack diagnosis. Initially, baseline results obtained using conventional machine learning classifiers were analyzed to establish a reference performance level. Subsequently, the impact of integrating metaheuristic optimization algorithms for feature selection was examined. The experimental analysis includes classification accuracy comparisons, confusion matrix evaluation, ROC curve analysis, convergence behavior of the optimization algorithm, and comparative assessment with existing studies reported in the literature.

Table 3. Performance metrics used to evaluate the proposed cardiovascular disease prediction framework.

Metric	Definition	Formula
Accuracy	The proportion of correctly classified instances among the total number of instances.	(TP + TN) / (TP + TN + FP + FN)
Precision	The proportion of correctly predicted positive cases among all predicted positive cases.	TP / (TP + FP)
Sensitivity (Recall)	The proportion of actual positive cases that are correctly identified by the model.	TP / (TP + FN)
Specificity	The proportion of actual negative cases that are correctly identified as negative.	TN / (TN + FP)
F1-score	The harmonic mean of precision and recall, providing a balanced measure of classification performance.	2 × (Precision × Recall) / (Precision + Recall)
ROC–AUC	The area under the Receiver Operating Characteristic curve, which evaluates the trade-off between the true positive rate and the false positive rate across different classification thresholds.	–

All experiments were implemented using the Python programming environment and executed on a workstation equipped with an Intel Core processor, 16 GB of RAM, and a Windows operating system. With respect to computational efficiency, all optimization algorithms were executed under identical hardware conditions using a fixed population size of 5 and 100 iterations. Under these settings, each algorithm completed the feature selection process within a practical runtime, with the JA exhibiting particularly favorable efficiency owing to its simple, parameter-free update mechanism. Detailed empirical runtime comparison across all algorithms will be investigated in future work. The machine learning models were implemented using the Scikit-learn library, which provides efficient implementations of classification algorithms and evaluation tools.

For the optimization-based feature selection stage, each metaheuristic algorithm was configured with a fixed population size and a predefined number of iterations to ensure a fair comparison among the evaluated optimization techniques. These parameter settings were selected based on commonly adopted configurations in metaheuristic optimization studies to provide a balanced trade-off between computational efficiency and search capability.

To evaluate the generalization capability of the proposed framework, each dataset was randomly divided into training and testing subsets using a 70:30 split ratio. This strategy enables the models to learn patterns from the training data while assessing their predictive performance on previously unseen samples. In addition, cross-validation techniques were considered during model evaluation to reduce sampling bias and improve the reliability and robustness of the obtained performance results.

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5.1 Cardiovascular Disease Dataset

The cardiovascular disease dataset used in this study was obtained from the Kaggle repository and consists of a large-scale collection of clinical and lifestyle information collected from 70,000 individuals for cardiovascular risk assessment. The dataset was originally designed to support predictive modeling and risk assessment for cardiovascular diseases. Each record corresponds to a unique patient and includes 11 input attributes together with a binary target variable indicating the presence or absence of cardiovascular disease.

The dataset contains a combination of demographic, physiological, and behavioral variables. The demographic attributes include age and gender, while physiological measurements include height, weight, systolic blood pressure, diastolic blood pressure, cholesterol level, and glucose level. In addition, the dataset incorporates lifestyle-related factors such as smoking status, alcohol consumption, and physical activity level. These attributes collectively capture a comprehensive set of factors known to influence cardiovascular health.

Due to its rich set of clinical and behavioral indicators, this dataset has been widely used in machine learning research related to cardiovascular disease prediction and risk classification. Consequently, it provides a reliable benchmark for evaluating predictive models in cardiovascular diagnosis tasks [29].

5.2 Heart Attack Dataset

The second dataset used in this study is a synthetic heart attack dataset that contains a diverse collection of cardiovascular health indicators and lifestyle attributes. The dataset includes patient-specific characteristics such as age, gender, cholesterol level, blood pressure, and heart rate, in addition to clinical indicators including diabetes status, family history of heart disease, smoking behavior, obesity status, and alcohol consumption.

Furthermore, the dataset incorporates lifestyle-related attributes such as physical activity duration, dietary habits, stress levels, and sedentary behavior patterns. Medical history information, including previous heart conditions, medication usage, and triglyceride levels, is also included. In addition to medical attributes, the dataset contains socioeconomic and geographical indicators such as income level, country, continent, and hemisphere, which provide additional contextual information that may influence cardiovascular risk patterns.

The dataset consists of 8,763 patient records collected worldwide and is labeled with a binary target variable indicating the presence or absence of heart attack risk. Owing to its diverse attribute set and global representation, this dataset serves as a valuable resource for evaluating machine learning models designed for cardiovascular risk prediction and diagnostic analysis [30, 38].

5.3 Result Analysis

To evaluate the effectiveness of the proposed framework, three machine learning NB, KNN, and DT classifiers were employed for cardiovascular disease detection and diagnosis. Initially, the baseline performance of these classifiers was evaluated without incorporating optimization-based feature selection. The baseline results for the two datasets are illustrated in Figures 6 and 7, which present the performance metrics obtained by conventional machine learning models.

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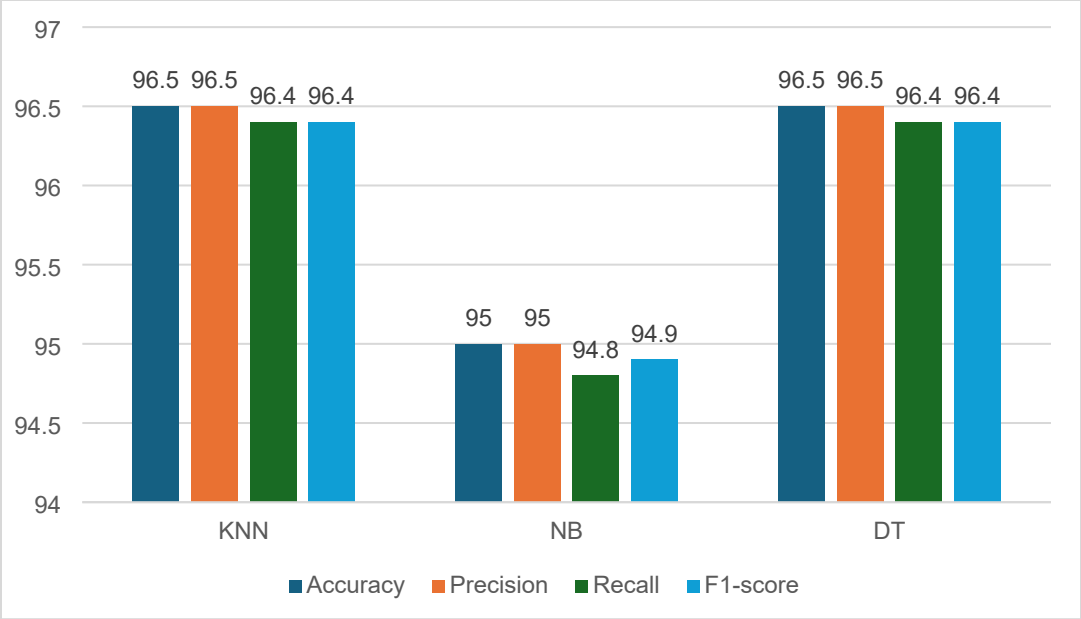


Figure 6. Baseline performance metrics of conventional machine learning classifiers (NB, KNN, and DT) on the cardiovascular disease dataset without optimization-based feature selection.

The obtained results indicate that although the individual classifiers provide reasonable baseline performance, their predictive accuracy remains limited when trained using the complete feature set. These limitations motivate the need for an enhanced framework capable of selecting the most informative features to improve classification performance.

To address this limitation, the proposed framework integrates several well-known JA, PSO, BA, FPA, FA, and WOA metaheuristic optimization techniques. These algorithms are employed to identify optimal subsets of features that maximize classification accuracy while reducing redundant information.

Table 4 presents the classification accuracy achieved by the three selected classifiers when combined with different optimization algorithms across the two datasets. The results demonstrate that the Jaya Algorithm (JA) consistently delivers the highest performance among the evaluated optimization methods. Integrating JA with the machine learning classifiers leads to notable improvements in classification accuracy compared with the other optimization strategies. These findings indicate that optimization-driven feature selection plays a critical role in enhancing classification performance and that the Jaya Algorithm is particularly effective in identifying informative feature subsets that contribute directly to improved predictive accuracy across different machine learning classifiers.

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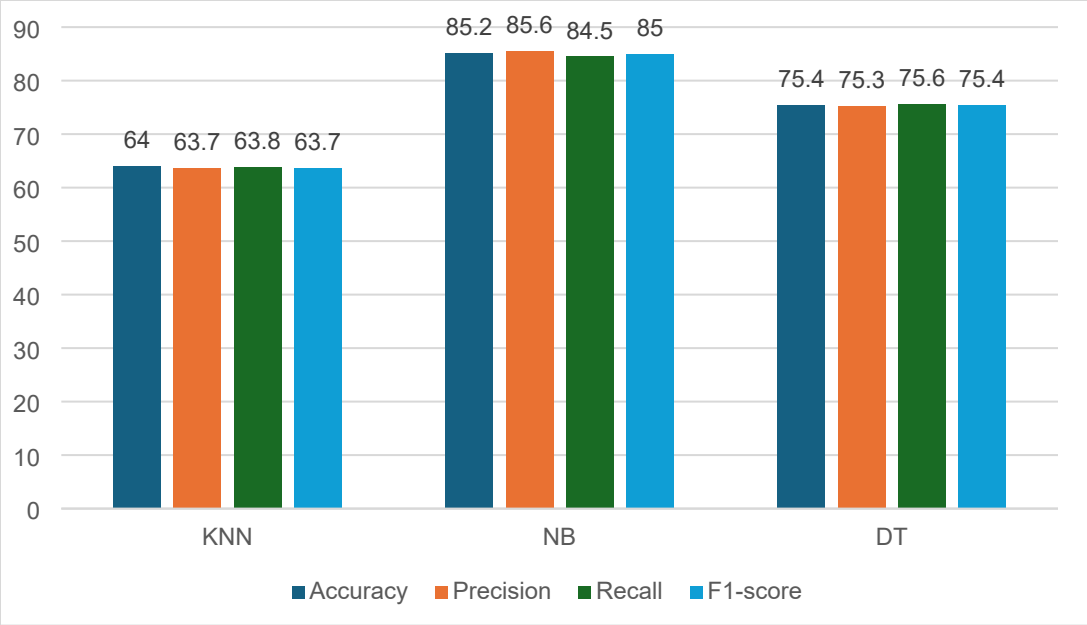


Figure 7. Baseline performance metrics of conventional machine learning classifiers (NB, KNN, and DT) on the heart attack dataset without optimization-based feature selection.

Table 4. Comparison of classification accuracy using features selected by different optimization algorithms.

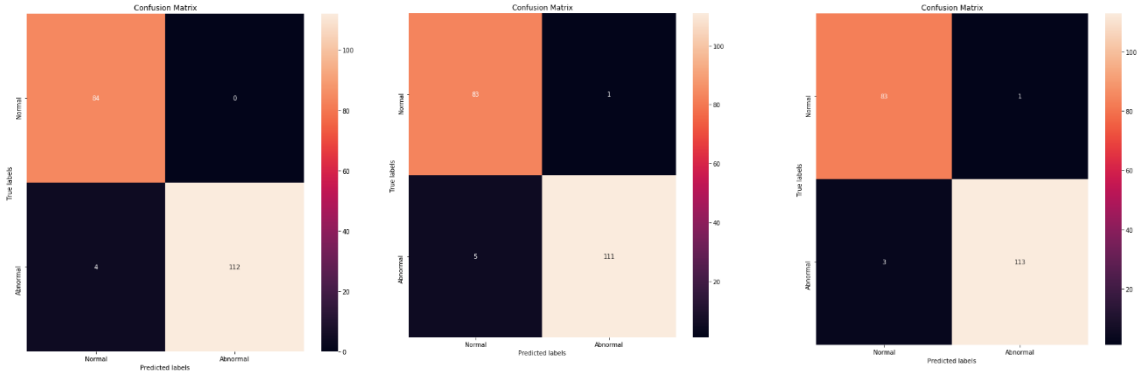
Model	Dataset	BA	WOA	PSO	FA	FPA	JA
KNN	Dataset A	97	97.5	97	97.5	97.5	98
	Dataset B	82	82	83.6	80.3	85.2	93.4
NB	Dataset A	96	96	96	95.5	96	97
	Dataset B	86.8	87	88.5	86.8	86.8	90
DT	Dataset A	97	97	97	97.5	97	98
	Dataset B	78.6	80	77	85.2	80.3	90.1

For Dataset A (CVD dataset), the JA-KNN and JA-DT models achieve the highest classification accuracy compared to other optimization-based approaches. In addition, the JA-NB classifier achieves competitive performance, attaining an accuracy of 97%. Similarly, for Dataset B (Heart attack dataset), improved performance is observed when optimization-based feature selection is applied. The JA-KNN model achieves the best performance among all classifiers and optimization algorithms, while the JA-DT model achieves the second-highest accuracy of 90.1%, followed by the JA-NB model. These results indicate that the integration of JA-based feature selection significantly enhances the predictive capability of machine learning classifiers for cardiovascular disease diagnosis.

To further analyze the classification performance of the proposed framework, the confusion matrices for the JA-based models are presented in Figures 8 and 9 for the two datasets. These matrices illustrate the distribution of correctly and incorrectly classified instances, enabling a detailed analysis of the classification behavior of each model.

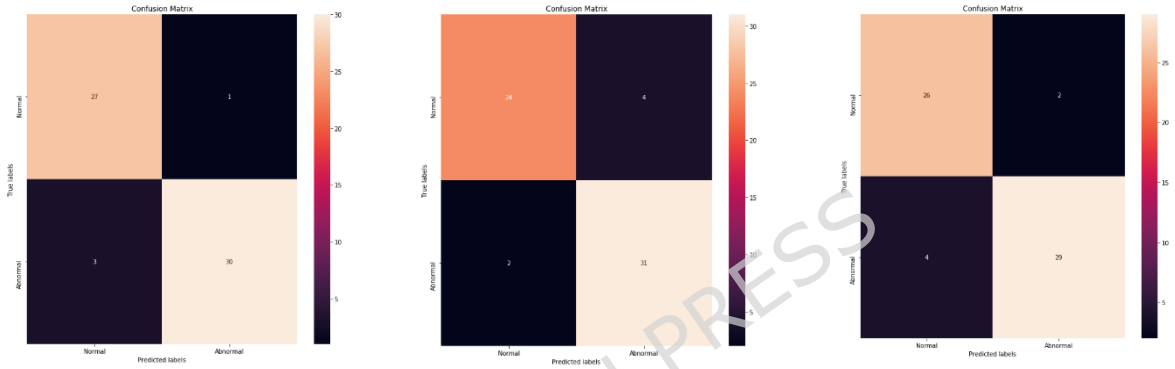
In addition, the ROC curves for the JA-based classifiers are shown in Figures 10 and 11, which illustrate the trade-off between the true positive rate and the false positive rate across different classification thresholds. The ROC analysis further confirms the superior performance of the JA-based models. Figure 12 presents a comparison between the classification accuracy of conventional machine learning models and the proposed JA-ML framework. The results clearly demonstrate that the proposed framework consistently achieves higher accuracy across all classifiers. In addition, the results clearly indicate that the integration of JA-based feature selection consistently improves the predictive performance of all classifiers compared with their conventional counterparts.

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JA-KNN **JA-NB** **JA-DT**

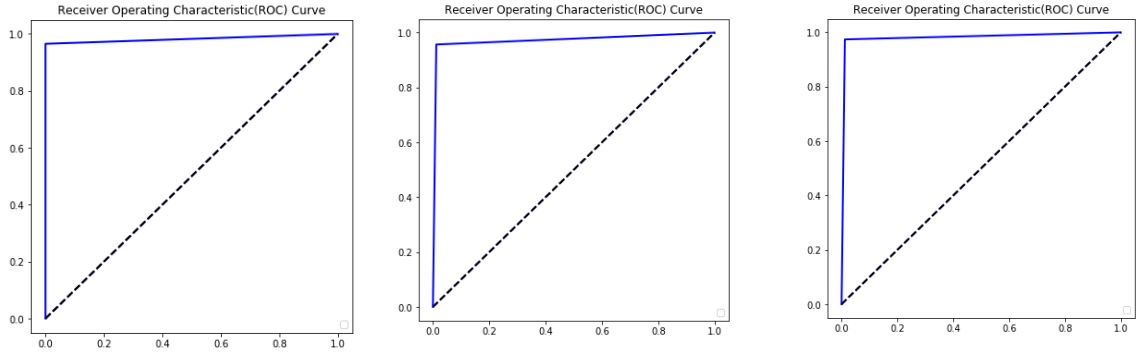
Figure 8. Confusion matrices of the JA-ML framework for the cardiovascular disease dataset using DT, NB, and KNN classifiers.



JA-KNN **JA-NB** **JA-DT**

Figure 9. Confusion matrices of the JA-ML framework for the heart attack dataset using DT, NB, and KNN classifiers.

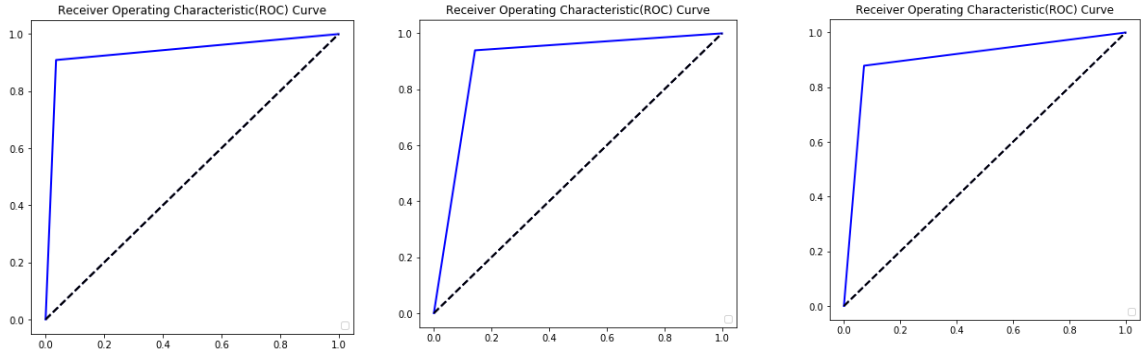
Furthermore, the comprehensive evaluation metrics of the JA-ML model are illustrated in Figures 13 and 14, which present precision, recall, and F1-score for both datasets. These metrics indicate that the proposed optimization-driven framework provides balanced classification performance across different evaluation criteria. Tables 5 and 6 present the class-wise performance of the JA-ML framework for both datasets, reporting precision, recall, and F1-score values for healthy and disease classes.



JA-KNN **JA-NB** **JA-DT**

Figure 10. ROC curves of the JA-ML framework for the cardiovascular disease dataset using DT, NB, and KNN classifiers.

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JA-KNN **JA-NB** **JA-DT**

Figure 11. ROC curves of the JA-ML framework for the heart attack dataset using DT, NB, and KNN classifiers.

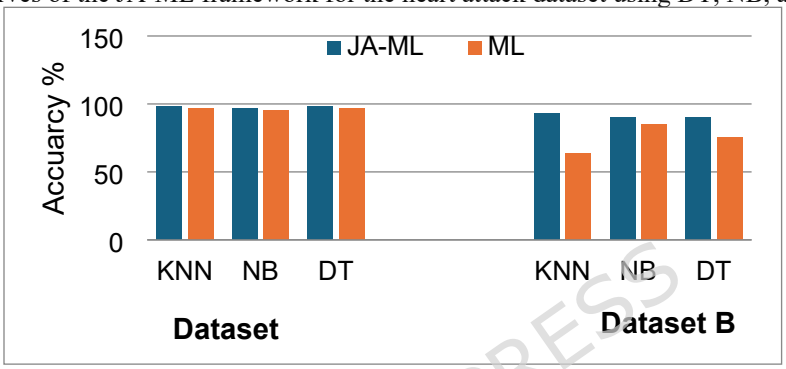


Figure 12. Comparison of classification accuracy between conventional machine learning classifiers and the proposed JA-ML optimization framework.

To further validate the contribution of the optimization-based feature selection stage, Figure 12 presents a direct comparison between the classification accuracy of conventional machine learning models trained on the full feature set (baseline) and those trained on the optimized feature subsets produced by the JA. The results consistently demonstrate that the integration of optimization-driven feature selection improves predictive performance across all three classifiers on both datasets, confirming the effectiveness of the proposed feature selection component within the overall framework.

In addition to classification performance evaluation, the convergence behavior of the proposed optimization framework was also analyzed. Figures 15 and 16 illustrate the convergence curves of the JA-based optimization process for both datasets, demonstrating the ability of the algorithm to efficiently converge toward optimal feature subsets during the search process. These results indicate that the optimization mechanism effectively balances exploration and exploitation, enabling stable and efficient identification of informative feature combinations.

Although the present study focused primarily on comparative evaluation using standard classification metrics, formal statistical significance analysis would provide stronger evidence regarding the reliability of the observed performance differences. Nonparametric tests such as the Wilcoxon signed-rank test and the Friedman test could be used to determine whether the performance improvements achieved by the JA-based models are statistically significant rather than attributable to random variation. Although such tests were not included in the current version of the study, all experiments were conducted under consistent preprocessing procedures, fixed hyperparameter settings, and identical evaluation conditions to ensure a fair comparison among the investigated methods. Future work

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will incorporate formal statistical significance analysis to further strengthen the rigor and robustness of the comparative evaluation.

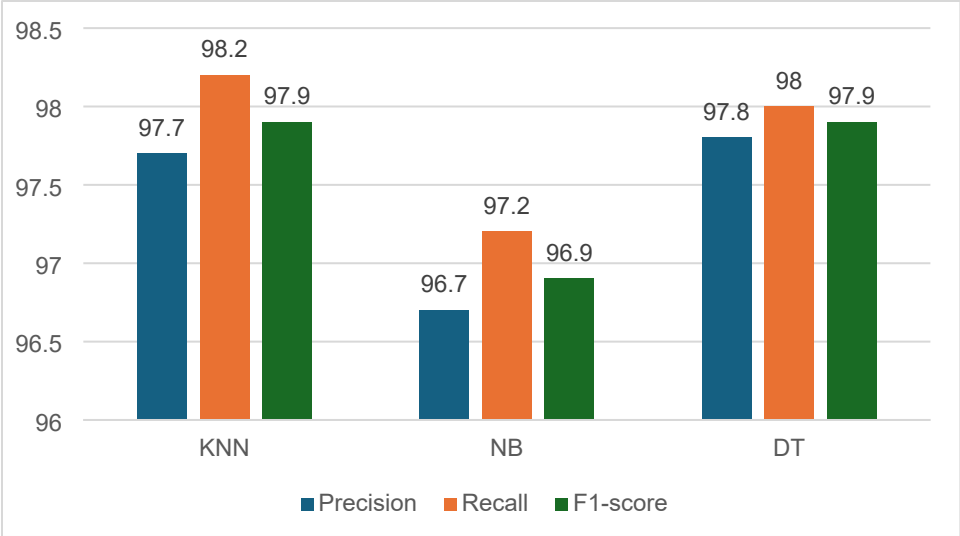


Figure 13. Overall evaluation metrics (precision, recall, and F1-score) of the JA-ML framework for the cardiovascular disease dataset.

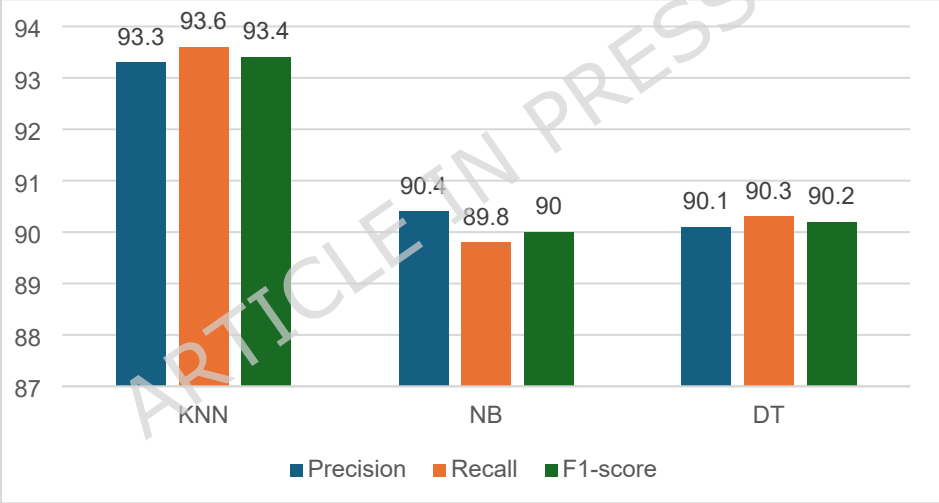


Figure 14. Overall evaluation metrics (precision, recall, and F1-score) of the JA-ML framework for the heart attack dataset.

Table 5. Class-wise performance of the proposed JA-ML model for Dataset A.

Model	Class	Precision	Recall	F1-score
JA-KNN	Healthy	95	100	98
JA-KNN	Disease	100	97	98
JA-NB	Healthy	94	99	97
JA-NB	Disease	99	96	97
JA-DT	Healthy	97	99	98
JA-DT	Disease	99	97	98

Table 6. Class-wise performance of the proposed JA-ML model for Dataset B.

Model	Class	Precision	Recall	F1-score
JA-KNN	Healthy	90	96	93
JA-KNN	Disease	97	91	94
JA-NB	Healthy	92	86	89
JA-NB	Disease	89	94	91
JA-DT	Healthy	87	93	90

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JA-DT	Disease	94	88	91

To complement the methodological overview presented in Table 1, Table 7 provides a direct quantitative comparison between the proposed framework and representative studies reported in the literature. The results indicate that the proposed JA-ML framework achieves superior predictive performance compared with previously reported approaches for both cardiovascular disease and heart attack prediction tasks. Whereas many existing studies rely on individual feature-selection techniques or limited hybrid strategies, the proposed framework systematically evaluates multiple metaheuristic optimization algorithms within a unified experimental environment. The superior performance achieved by the JA-based feature-selection strategy demonstrates its effectiveness in identifying informative clinical attributes that enhance the predictive capability of machine learning models for cardiovascular disease diagnosis. These findings confirm that integrating metaheuristic optimization with machine learning classifiers provides a robust and effective approach for improving predictive accuracy in intelligent healthcare systems.

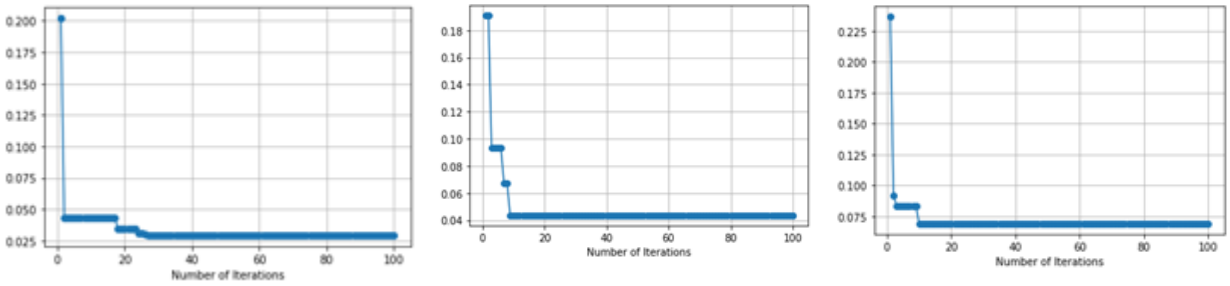
The superior performance of the Jaya Algorithm may be attributed to its simple yet effective optimization mechanism. Unlike several other metaheuristic algorithms, JA does not rely on algorithm-specific control parameters, which reduces the need for extensive parameter tuning and lowers the risk of suboptimal configuration settings. Instead, the algorithm iteratively updates candidate solutions by simultaneously guiding them toward the best solution and away from the worst solution within the population. This search strategy promotes an effective balance between exploration and exploitation, supports stable convergence behavior, and facilitates the identification of informative feature subsets. These characteristics likely contributed to the consistently improved classification performance achieved by the JA-based models across the evaluated cardiovascular datasets. Moreover, the parameter-free nature of JA makes it particularly suitable for comparative experimental settings, as it reduces sensitivity to manual tuning and enables a more consistent assessment of feature-selection effectiveness across datasets and classifiers.

Although computational cost is an important consideration when evaluating optimization algorithms, the primary objective of the present study was to improve prediction performance for cardiovascular disease detection within a smart healthcare environment. In such applications, predictive accuracy is a critical factor because inaccurate predictions may negatively affect clinical decision-making and patient outcomes. Nevertheless, computational efficiency remains an important practical consideration, particularly for real-time or large-scale deployment. In this context, the Jaya Algorithm exhibited a favorable balance between predictive performance and computational efficiency. Its simple update mechanism and parameter-free design reduce the need for extensive algorithm-specific tuning and lower implementation complexity compared with several other methods. These characteristics, together with its strong predictive performance, make JA a practically attractive optimization method for prediction for cardiovascular disease. A more detailed empirical runtime comparison and formal computational complexity analysis of the investigated optimization algorithms will be considered in future work.

In addition to the quantitative performance gains, the selected feature subsets were examined qualitatively to assess their alignment with established cardiovascular risk characteristics. Beyond the observed improvements in classification accuracy, the feature subsets identified by the optimization algorithms may also provide useful analytical insight into cardiovascular disease prediction. Several frequently selected attributes correspond to well-established cardiovascular risk factors commonly considered in clinical assessment, suggesting that the proposed framework captures medically meaningful patterns in the data. The repeated selection of these attributes across different

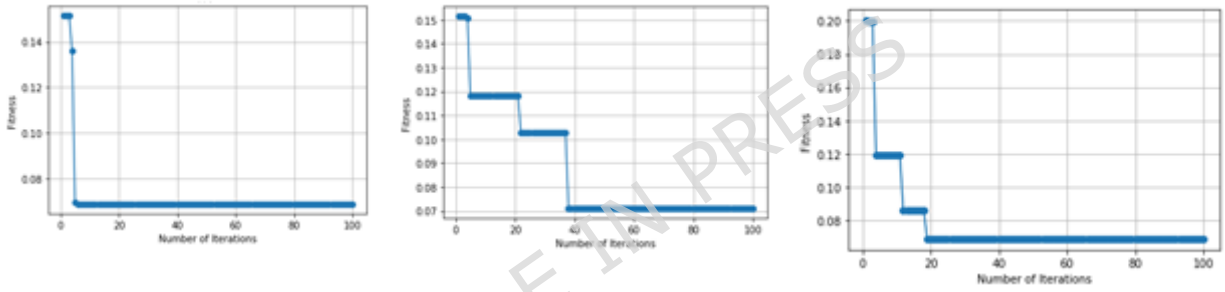
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optimization algorithms further supports their relevance to disease prediction and strengthens the practical applicability of the proposed approach. Nevertheless, the primary objective of this study was to conduct a comparative evaluation of metaheuristic optimization algorithms and machine learning models rather than a detailed clinical interpretation of individual risk factors. Accordingly, future work may involve collaboration with medical experts to further investigate the clinical significance, interpretability, and decision-support value of the selected features.



JA-KNN **JA-NB** **JA-DT**

Figure 15. Convergence curves of the JA-based optimization process for the cardiovascular disease dataset using DT, NB, and KNN classifiers.



JA-KNN **JA-NB** **JA-DT**

Figure 16. Convergence curves of the JA-based optimization process for the heart attack dataset using DT, NB, and KNN classifiers.

Table 7. Comparison between the proposed JA-ML framework and previous studies.

Study	Dataset	Classifier	Results
Mohan et al. [31]	CVD	HRFLM	Accuracy: 88%
Singh et al. [32]	CVD	KNN / LR / DT	KNN: 87%
Gavhane et al. [33]	CVD	Neural Network	Precision: 91%, Recall: 89%
Kavitha et al. [34]	CVD	RF + DT	Accuracy: 88%
Sharma et al. [35]	Heart Attack	RF / DT / LR	Accuracy: 92.1%
Atallah and Mousa [36]	Heart Attack	ML models	Accuracy: 90%
Nesreen [37]	Heart Attack	Neural Network	Accuracy: 89.3%
Proposed Framework	CVD	JA-ML	Accuracy: 98%
Proposed Framework	Heart Attack	JA-ML	Accuracy: 93.4%

6. Limitations and Future Work

Although the proposed hybrid OT–ML framework demonstrated promising performance for cardiovascular disease prediction, several limitations should be acknowledged. First, the current study evaluated the framework using only three machine learning classifiers, namely NB, KNN, and DT. These models were intentionally selected to provide a clear, interpretable, and computationally efficient assessment of the impact of metaheuristic feature-selection algorithms within a smart healthcare setting. Nevertheless, incorporating more advanced learning algorithms, such as Support Vector Machines (SVM), Random Forest (RF), XGBoost, deep learning architectures, and ensemble

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strategies including voting, stacking, and boosting, could provide deeper insight into predictive performance, robustness, and generalization capability.

Second, the present study primarily focused on classification accuracy and related evaluation metrics. Although these measures provide a useful basis for comparative assessment, future research should incorporate additional evaluation strategies, including statistical significance testing, model robustness analysis, and cross-dataset validation, to provide a more comprehensive assessment of the proposed framework. In particular, formal statistical tests such as the Wilcoxon signed-rank test and the Friedman test may be employed to determine whether the observed performance differences among optimization algorithms and classifiers are statistically significant.

Third, although multiple metaheuristic optimization algorithms were investigated, the proposed framework relied on standard implementations of these methods. Future work may therefore explore modified or hybrid optimization strategies, such as adaptive variants of the Jaya Algorithm or hybrid metaheuristic approaches, to further improve feature-selection efficiency and overall model performance.

Another important direction for future research involves evaluating the proposed framework on larger and more diverse medical datasets. Investigating its performance across different disease categories, heterogeneous clinical data sources, and multi-center healthcare environments would enable a more thorough assessment of its scalability, robustness, and generalizability in real-world applications.

Finally, future developments may focus on enhancing the proposed smart healthcare system through the integration of additional intelligent components, such as explainable artificial intelligence techniques, wearable sensor-based real-time patient monitoring, and cloud-based medical analytics platforms. Such extensions could further improve the reliability, interpretability, and clinical applicability of the proposed diagnostic framework.

7. Conclusion

This study presented a hybrid healthcare framework that integrated metaheuristic optimization techniques with machine learning models for cardiovascular disease detection and prediction. The proposed framework combined optimization-driven feature selection with machine learning classification to improve diagnostic accuracy and identify the most informative clinical attributes associated with cardiovascular risk. To evaluate the effectiveness of the proposed framework, two datasets related to cardiovascular disease and heart attack prediction were analyzed. Six metaheuristic optimization algorithms, namely the JA, PSO, BA, FPA, FA, and WOA, were employed to identify optimal feature subsets. The selected features were subsequently used to train three machine learning classifiers: NB, KNN, and DT. The experimental results demonstrated that integrating optimization-based feature selection significantly improved classification performance compared with conventional machine learning models trained on the full feature set. Among the evaluated optimization methods, the Jaya Algorithm consistently achieved the best overall performance across both datasets. In particular, the hybrid JA–KNN and JA–DT models achieved the highest classification accuracy on the cardiovascular disease dataset, reaching 98%. For the heart attack dataset, the JA–KNN model achieved the best performance with an accuracy of 93.4%, followed by the JA–DT model with an accuracy of 90.1%. These findings indicated that the integration of metaheuristic optimization techniques with machine learning classifiers provided an effective strategy for enhancing predictive performance in cardiovascular disease diagnosis. Furthermore, the proposed framework demonstrated strong potential for deployment in smart healthcare

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environments, particularly when integrated with mobile applications and cloud-based medical monitoring systems. The generalization capability of the proposed framework was supported by its evaluation on two independent cardiovascular datasets, where it demonstrated consistent predictive performance across both datasets. This consistency suggested that the proposed optimization–machine learning framework could capture relevant patterns associated with cardiovascular disease prediction beyond a single data source. Nevertheless, variations in patient demographics, clinical practices, data quality, and healthcare infrastructures may influence model performance in real-world settings. Therefore, further validation using larger, more diverse, and multi-center cardiovascular datasets is recommended to more comprehensively assess the robustness and generalizability of the proposed framework across different clinical environments.

Declarations

Conflict of Interest Statement: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Data Availability: The datasets used in this study are publicly available on Kaggle. The Cardiovascular Disease Dataset is available at: https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset. The Heart Attack Dataset is available at: https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset. These datasets were accessed for the purposes of this study and are publicly accessible to researchers.

Authors' Contributions: Samar M. Zayed significantly contributed to methodology refinement, dataset curation, provided technical results, and critically reviewed and revised the manuscript. Samar M. Zayed also created Figures 3 and 5. Walid El-Shafai managed project administration, validation of experimental results, data analysis, supervised the research process, performed the final editing, and acted as the corresponding author. Hanan Khalid A Aljasim significantly contributed to dataset curation, provided technical results, and critically reviewed and revised the manuscript. Mai Alduailij significantly contributed to methodology refinement, provided technical results, and critically reviewed and revised the manuscript, Fatma Khallaf was responsible for the conceptualization of the study, developing the methodology, and preparing the original manuscript draft.

Code Availability: The implementation code used for data preprocessing, metaheuristic feature selection, machine learning classification, and performance evaluation is not publicly available because it forms part of an ongoing institutional research project. However, the relevant code may be made available by the corresponding author upon reasonable request and subject to appropriate conditions of use.

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